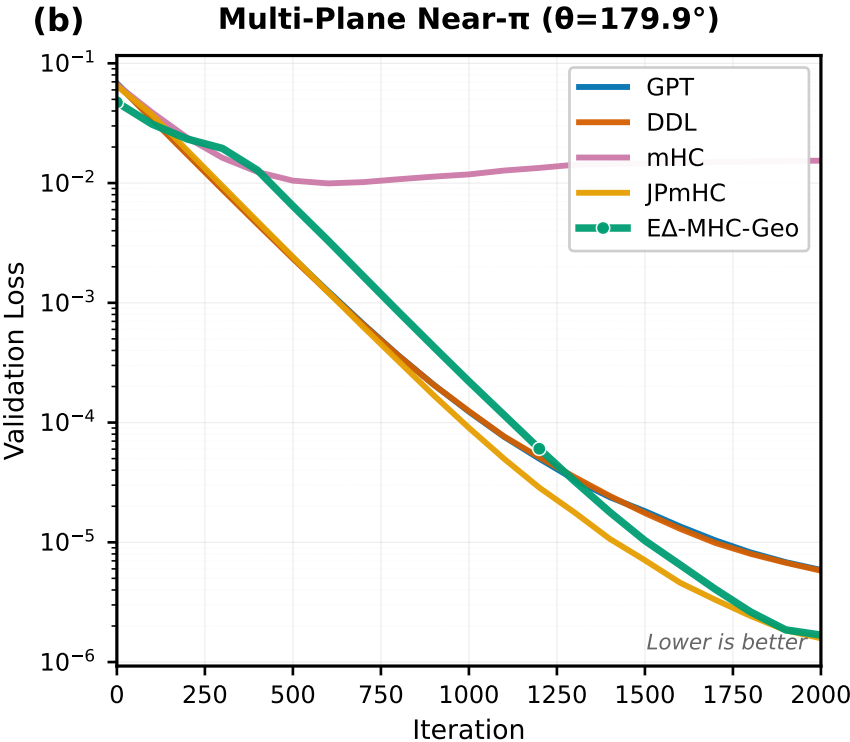
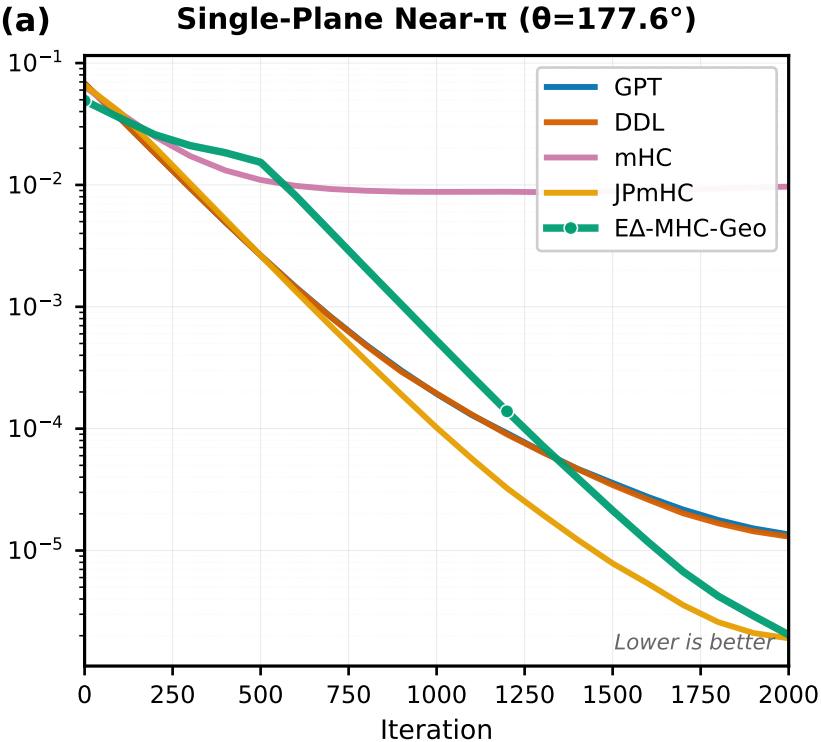


Near- π Rotation Analysis: All Geometric Models Excel on Pure Rotation



— Performance Summary —

Final Validation Loss:

Model	Single-Pl	Multi-Pl
GPT	$1.20\text{e-}05$	$4.93\text{e-}06$
DDL	$1.16\text{e-}05$	$4.85\text{e-}06$
mHC	$9.83\text{e-}03$	$1.55\text{e-}02$
JPmHC	$1.70\text{e-}06$	$1.32\text{e-}06$
EΔ	$1.16\text{e-}06$	$1.51\text{e-}06$

All $S0(n)$ models excel at near- π rotation (no negation).

